Cycle Gravity Spectra and the Sharp Cycle Classification for Graffiti 197

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Research edition: 18 October 2026; submission revision

Abstract. Roucairol and Cazenave refuted Graffiti Conjecture 197 using the cycle C_{17} and conjectured an infinite family of further counterexamples at orders $4r+1 \geq 17$. Under their numerical-spectral-spread convention, we give a self-contained proof that *every* cycle C_n with $n \geq 15$ violates the inequality, while every C_n with $3 \leq n \leq 14$ satisfies it. The key is an exact positive-semidefinite decomposition of the shifted reciprocal-distance matrix into cyclic-interval Gram matrices, yielding a uniform $-3/2$ lower spectral bound. The finite complement is certified by a small exact-rational Fourier interval calculation, with a separate elementary witness at C_{14}. We also establish a general Gram certificate for nonincreasing discretely convex radial kernels on cycles, including reciprocal powers d^{-p} for $p > 0$. Existing results on generalized distance spectra are distinguished from the claims under consideration; historical originality is not asserted. Lean 4 formalization has certified the matrix and harmonic input identities but not the entire final graph-spectral assertion.

Keywords. Spectral graph theory; cycle graph; Graffiti conjectures; Harary matrix; positive semidefinite decomposition; convex distance kernels; exact computation.

1 The Graffiti question and attribution

Graffiti and its later computational study generated many conjectural graph-invariant inequalities [1, 2, 3]. Roucairol and Cazenave [4, §5.1–5.2] exhibited C_{17} as a counterexample to Graffiti 197, also reported C_{21} and C_{25}, and conjectured the entire family C_{4r+1} for $r \geq 4$. Their use of "range" is potentially ambiguous: here it means *maximum eigenvalue minus minimum eigenvalue*, not the count of distinct values.

For the simple cycle C_n on $\mathbb{Z}/n\mathbb{Z}$, let A_n be its adjacency matrix, $d(u,v)$ its graph distance, and R_n the zero-diagonal Harary matrix with $(R_n)_{uv} = 1/d(u,v)$ for $u \neq v$. The gravity normalization of [4] specializes, since all degrees are two, to

$$(\mathsf{Gr}_n)_{uv} = \begin{cases} \dfrac{4}{(n-1)d(u,v)}, & u \neq v, \\ 0, & u = v, \end{cases} \tag{1}$$

Thus $\mathsf{Gr}_n = \frac{4}{n-1} R_n$. For a real symmetric matrix write $\operatorname{spr}(M) = \lambda_{\max}(M) - \lambda_{\min}(M)$, and order adjacency eigenvalues with multiplicity as $\lambda_1(A_n) \geq \cdots \geq \lambda_n(A_n)$. The cycle restriction of Graffiti 197 is

$$-\lambda_{n-1}(A_n) \leq \operatorname{spr}(\mathsf{Gr}_n). \tag{2}$$

We establish the following sharp, cycle-specific classification.

*Newton Astro Labs is the trading name under which Tony Newton conducts independent computational research in the United Kingdom; it is not a limited company.

Theorem 1 (Sharp cycle threshold)**.** *For every integer $n \geq 3$, the inequality* (2) *fails strictly if and only if $n \geq 15$.*

Theorem 1 is a stronger family result than the published examples and conjecture; it is *not* a claim to have first refuted Graffiti 197. The proof and finite certificate below are mathematical claims for review; priority is a separate question.

2 An exact positive-semidefinite identity

Put $m = \lfloor n/2 \rfloor$ and let J be the all-ones matrix. For $2 \leq j \leq m$, let B_j be the circulant zero-one matrix whose v-th column records the cyclic interval $\{v, v+1, \ldots, v+j-1\}$. Since $2j \leq n$, the number of common elements in intervals separated by distance d is $(j-d)_+$, hence

$$(B_j B_j^{\mathsf{T}})_{uv} = (j - d(u,v))_+. \tag{3}$$

Define the strictly positive coefficients

$$w_j = \begin{cases} \dfrac{2}{(j-1)j(j+1)}, & 2 \leq j < m, \\ \dfrac{1}{m(m-1)}, & j = m. \end{cases} \tag{4}$$

Lemma 1 (Cyclic-interval certificate)**.** *For every cycle C_n with $n \geq 4$,*

$$R_n + \frac{3}{2} I = \frac{1}{m} J + \sum_{j=2}^{m} w_j B_j B_j^{\mathsf{T}} \succeq 0. \tag{5}$$

In particular $\lambda_{\min}(R_n) \geq -3/2$.

Proof. For $1 \leq d \leq m$ discrete summation by parts gives

$$\frac{1}{d} = \frac{1}{m} + \sum_{j=2}^{m} w_j (j-d)_+. \tag{6}$$

For $j < m$, $w_j = (j-1)^{-1} - 2j^{-1} + (j+1)^{-1}$; at the endpoint $w_m = (m-1)^{-1} - m^{-1}$. Thus (6), together with (3), supplies all off-diagonal entries of (5). At $d = 0$ the right-hand diagonal is $\frac{1}{m} + \sum_{j=2}^{m} j w_j = \frac{3}{2}$, including $m = 2$. Each $B_j B_j^{\mathsf{T}}$ and J is a Gram matrix and every coefficient is nonnegative. This proves the matrix identity and its Loewner consequence. □

3 The uniform infinite-family proof

The nonnegative symmetric circulant matrix R_n has constant row sum s_n, which is its largest eigenvalue. By counting the two vertices at each non-antipodal distance and the single antipode when n is even,

$$s_{2m+1} = 2H_m, \qquad s_{2m} = 2H_{m-1} + \frac{1}{m}, \tag{7}$$

where $H_m = \sum_{j=1}^{m} 1/j$. Lemma 1 consequently yields

$$\operatorname{spr}(\mathsf{Gr}_n) \le \frac{4}{n-1}\left(s_n + \frac{3}{2}\right). \tag{8}$$

The cycle adjacency spectrum, with multiplicity, is $\{2\cos(2\pi k/n) : 0 \le k < n\}$. Sorting it gives

$$-\lambda_{n-1}(A_n) = \begin{cases} 2\cos\left(\frac{\pi}{2m+1}\right), & n = 2m+1, \\ 2\cos\left(\frac{\pi}{m}\right), & n = 2m. \end{cases} \tag{9}$$

For odd cycles with $m \ge 7$, the gravity upper bound in (8) is $q_o(m) = (4H_m + 3)/m$. It decreases strictly as m increases, whereas $2\cos(\pi/(2m+1))$ increases. At the boundary $m = 7$,

$$2\cos(\pi/15) - q_o(7) > 2 - \left(\frac{22}{105}\right)^2 - \frac{468}{245} = \frac{506}{11025} > 0. \tag{10}$$

We used $\cos x \ge 1 - x^2/2$ and $\pi < 22/7$. This proves strict failure of (2) for every odd $n \ge 15$.

For even cycles with $m \ge 8$, write $q_e(m) = 4(2H_{m-1} + 1/m + 3/2)/(2m-1)$. Put $S_m = 2H_{m-1} + 1/m + 3/2$. Then $S_{m+1} - S_m = 1/m + 1/(m+1)$, and $(2m-1)(S_{m+1} - S_m) < 4 < 2S_m$ for $m \ge 2$, proving $q_e(m) > q_e(m+1)$. Moreover $2\cos(\pi/m)$ is increasing. The boundary comparison is

$$2\cos(\pi/8) - q_e(8) > 2 - \left(\frac{11}{28}\right)^2 - \frac{1907}{1050} = \frac{1733}{58800} > 0. \tag{11}$$

This establishes the $n \ge 15$ implication of Theorem 1 in both parities without asymptotics or a numerical eigensolver.

4 A sharp finite complement

For $3 \le n \le 14$, let $k = \lfloor n/2 \rfloor$. The Fourier vector at frequency k is an exact eigenvector of R_n; denote its eigenvalue by

$$\eta_{n,k} = 2\sum_{d=1}^{\lfloor (n-1)/2 \rfloor} \frac{\cos(2\pi k d/n)}{d} + \mathbf{1}_{2|n}\frac{(-1)^k}{n/2}. \tag{12}$$

Since $s_n = \lambda_{\max}(R_n)$ and $\lambda_{\min}(R_n) \le \eta_{n,k}$,

$$\operatorname{spr}(\mathsf{Gr}_n) \ge \frac{4}{n-1}(s_n - \eta_{n,k}). \tag{13}$$

It therefore suffices to compare the exact right side of (13) against (9) for the twelve orders. The companion `exact_certificate.py`, using standard-library rational arithmetic only, supplies strict lower enclosures for (13) and strict upper enclosures for (9). Its acceptance path uses neither binary floating point nor approximations without error bounds. The complete table and construction are in the supplemental verification note.

An especially short independent certificate for the boundary C_{14} is worth recording because it already appeared in the earlier two-page note. The constant Fourier mode of R_{14} has eigenvalue $353/70$ and the alternating mode has eigenvalue $-289/210$. Thus

$$\operatorname{spr}(\mathsf{Gr}_{14}) \ge \frac{4}{13}\left(\frac{353}{70} + \frac{289}{210}\right) = \frac{2696}{1365} > \frac{15}{8}, \tag{14}$$

while

$$-\lambda_{13}(A_{14}) = 2\cos(\pi/7) < 2\cos(\pi/8) = \sqrt{2+\sqrt{2}} < \frac{15}{8}.$$

Combining the twelve exact-rational comparisons with the preceding infinite-family argument proves Theorem 1.

5 A convex radial-kernel extension

DeVille [5] established a general framework for graph matrices whose entries depend on distance, including spectral analyses on distance-regular graphs. The following cyclic-interval Gram representation is a related constructive sufficient condition for a spectral lower bound. It is included here as a by-product of the proof above; we do not assert that the general statement is historically new.

Theorem 2 (Discrete-convex cyclic Gram certificate)**.** *Fix $n \ge 4$ and $m = \lfloor n/2 \rfloor$. Let $f_1 \ge f_2 \ge \cdots \ge f_m \ge 0$ satisfy $f_{j-1} - 2f_j + f_{j+1} \ge 0$ for $2 \le j < m$. Put $\delta_j = f_{j-1} - 2f_j + f_{j+1}$, and let $(K_f)_{uv} = f_{d(u,v)}$ off the diagonal and zero on the diagonal. Then*

$$K_f + (2f_1 - f_2)I = f_m J + \sum_{j=2}^{m-1} \delta_j B_j B_j^{\mathsf{T}} + (f_{m-1} - f_m) B_m B_m^{\mathsf{T}} \succeq 0. \tag{15}$$

Consequently $\lambda_{\min}(K_f) \ge -(2f_1 - f_2)$. The additive shift is optimal as a uniform bound over all $n \ge 4$: equality holds for C_4.

Proof. All coefficients on the right of (15) are nonnegative by assumption. For $1 \le d \le m$, discrete telescoping of the differences $f_j - f_{j+1}$ gives

$$f_d = f_m + \sum_{j=2}^{m-1} \delta_j (j-d)_+ + (f_{m-1} - f_m)(m-d)_+.$$

Equation (3) verifies the off-diagonal matrix entries. On the diagonal the same right-hand scalar is $2f_1 - f_2$ by telescoping. At $n = 4$ the alternating vector $(1, -1, 1, -1)$ is an eigenvector of K_f with eigenvalue $-2f_1 + f_2$, proving the uniform sharpness assertion. □

Corollary 1 (Reciprocal powers)**.** *For $p > 0$, let $(K_{n,p})_{uv} = d(u, v)^{-p}$ when $u \neq v$ and zero otherwise. Then for every $n \geq 4$,*

$$\lambda_{\min}(K_{n,p}) \geq -\big(2 - 2^{-p}\big).$$

Moreover, for the modified *matrices $\mathsf{Gr}_{n,p} = 4K_{n,p}/(n-1)$, one has $\operatorname{spr}(\mathsf{Gr}_{n,p}) \to 0$ as $n \to \infty$, so the reverse of* (2) *eventually holds if Gr_n is replaced by $\mathsf{Gr}_{n,p}$.*

Proof. The sequence $f_d = d^{-p}$ is decreasing and discretely convex, so Theorem 2 applies. Its row sum is at most $2\sum_{d=1}^{\lfloor n/2 \rfloor} d^{-p}$ and is the largest eigenvalue. The latter is $O(n^{1-p})$ for $0 < p < 1$, $O(\log n)$ for $p = 1$, and $O(1)$ for $p > 1$. In each case multiplying the row-sum-plus-fixed-shift spread bound by $4/(n-1)$ gives $o(1)$. Equation (9) tends to 2. □

Remark 1. *The original Graffiti 197 statement uses $p = 1$. The $p \neq 1$ statement is a mathematical generalization of the distance kernel, not a claim to refute any additional named graph conjecture.*

6 Reproducibility and certification boundaries

The companion exact-rational verifier proves the C_3–C_{14} finite complement, and checks (15) for $n = 4, \ldots, 34$ and $p \in \{1, 2, 3\}$ (41,013 exact matrix-entry evaluations). The finite comparison is a computer-assisted *exact* certificate, whereas the $n \geq 15$ proof and Theorem 2 are algebraic/analytic. Proof scripts, certificate JSON, checksums, and a stage-status ledger are supplied separately.

The project's Lean issue #12 documents a 15-stage formalization. GitHub Actions run 39160177061 succeeded (7,376 build jobs, 301 declarations audited). In Lean 4/mathlib this includes the exact cyclic Gram matrix identities, two-sided Loewner inequalities for the actual gravity matrix, cyclic distance-shell cardinalities, and the parity-specific harmonic row sums. *The full graph-spectral classification has not been compiled as an unconditional Lean theorem.* Remaining are the final real spectral-span extraction, ordered second-smallest adjacency identification including multiplicity, and the finite complementary Lean proofs. These tasks should not be confused with gaps in the paper's ordinary written derivation.

The original Graffiti 197 refutation belongs to Roucairol and Cazenave. Generalized graph distance spectra and reciprocal-distance Laplacian methods have an established literature [5, 6]; no independent priority certificate for the present sharpening or convex-kernel form is claimed. The analysis is submitted for independent mathematical and bibliographic assessment.

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